Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.128860483425, median 0.109725894339, std: 0.0919416482561
Reprojection error (cam1): mean 0.132964547229, median 0.114340829682, std: 0.0922496498981
Gyroscope error (imu0): mean 0.465047123491, median 0.301071903886, std: 0.541327356966
Accelerometer error (imu0): mean 0.276352255797, median 0.219924731838, std: 0.195597875674

Residuals

Reprojection error (cam0) [px]: mean 0.128860483425, median 0.109725894339, std: 0.0919416482561
Reprojection error (cam1) [px]: mean 0.132964547229, median 0.114340829682, std: 0.0922496498981
Gyroscope error (imu0) [rad/s]: mean 0.0131535189837, median 0.00851559939451, std: 0.0153110497981
Accelerometer error (imu0) [m/s^2]: mean 0.078164221628, median 0.0622041076933, std: 0.0553234337098

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.10268794 -0.99340566 0.05099384 0.00540797]
[0.0252462 -0.0486455 -0.99849699 -0.02606844]
[0.99439319 0.103821   0.02008442 -0.02177582]
[0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[0.10268794 0.0252462   0.99439319 0.02175653]
[-0.99340566 -0.0486455   0.103821   0.00636499]
[0.05099384 -0.99849699 0.02008442 -0.02586768]
[0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.0227186866424

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.10324856 -0.99335737 0.0508022 -0.04470342]
[0.02407144 -0.04856477 -0.99852993 -0.02608966]
[0.99436427 0.10431966 0.0188973 -0.02177516]
[0.   0.   0.   1.   ]]

T_ic: (cam1 to imu0):
[[0.10324856 0.02407144 0.99436427 0.02689602]
[-0.99335737 -0.04856477 0.10431966 -0.04340193]
[0.0508022 -0.99852993 0.0188973 -0.02336878]
[0.   0.   0.   1.   ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.0227131135717

Baselines:

Baseline (cam0 to cam1):
[[0.99999982 0.00020316 0.00055863 -0.05009393]
[-0.00020251 0.99999931 -0.00116046 -0.00004541]
[-0.00055887 0.00116035 0.99999917 0.00003392]
[0.   0.   0.   1.   ]]
baseline norm: 0.0500939639159 [m]

Gravity vector in target coords: [m/s^2]
[0.01550138 -9.73926172 -1.14671871]

Calibration configuration
=========================

cam0

Camera model: pinhole
Focal length: [373.2046890633468, 373.99920139935665]
Principal point: [319.29362305425343, 234.57418040343345]
Distortion model: radtan
Distortion coefficients: [-0.003870579707859219, -0.007933170708090008, -0.0012013485708556228, 0.0010482017949853587]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.0264 [m]

cam1

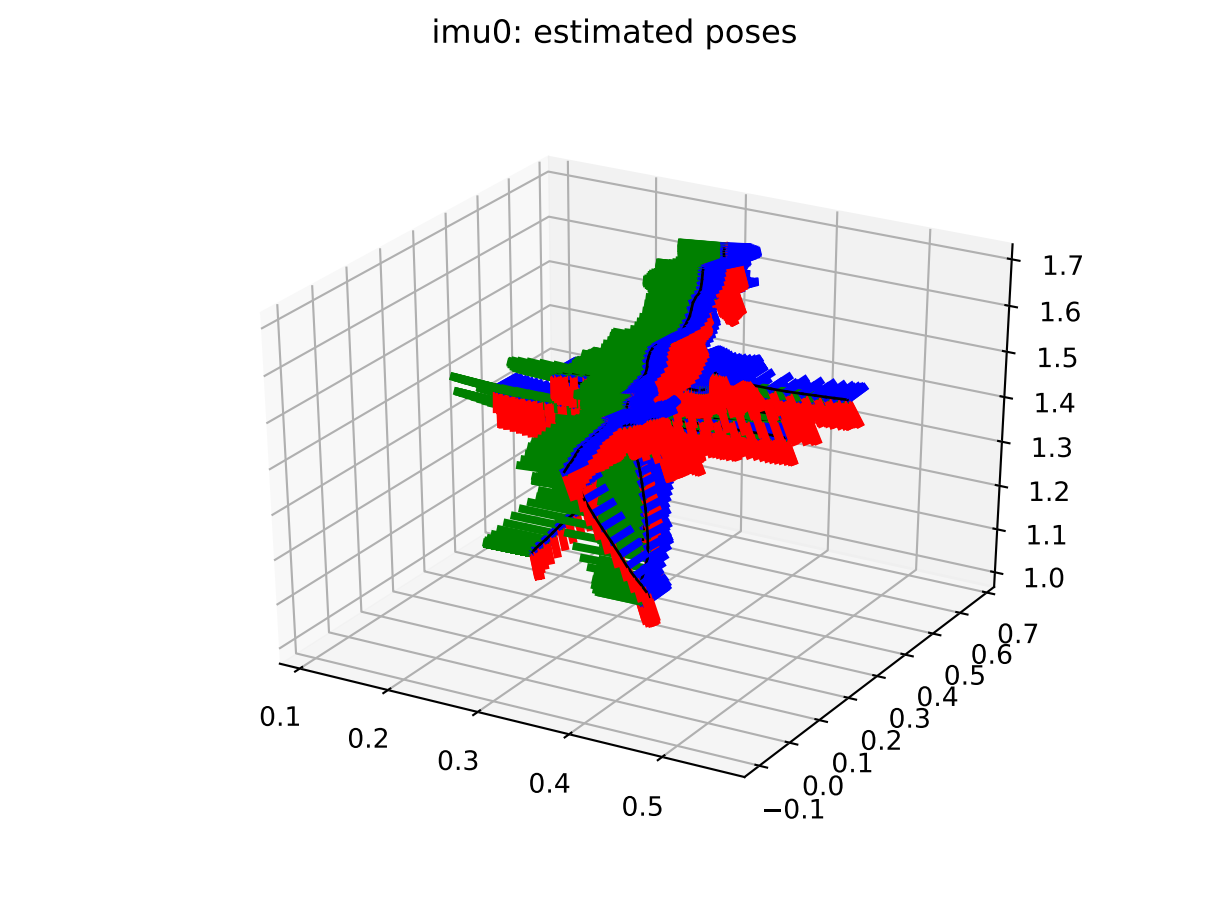
Camera model: pinhole
Focal length: [373.44427468507405, 374.1856416223415]
Principal point: [319.5149435679406, 235.42994355593646]
Distortion model: radtan
Distortion coefficients: [-0.003346859272076343, -0.009247214893833011, -0.001309957153721866, 0.0006674510320810162]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.0264 [m]

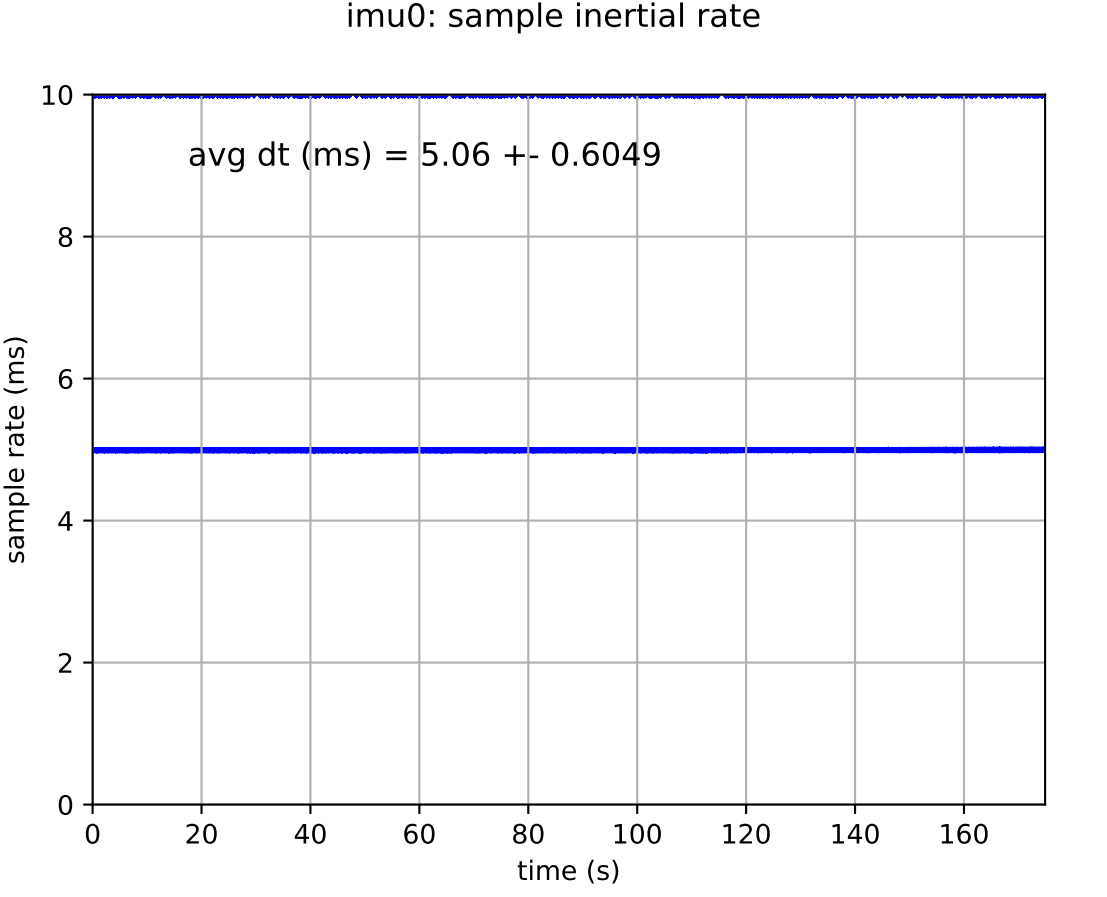
IMU configuration
==========

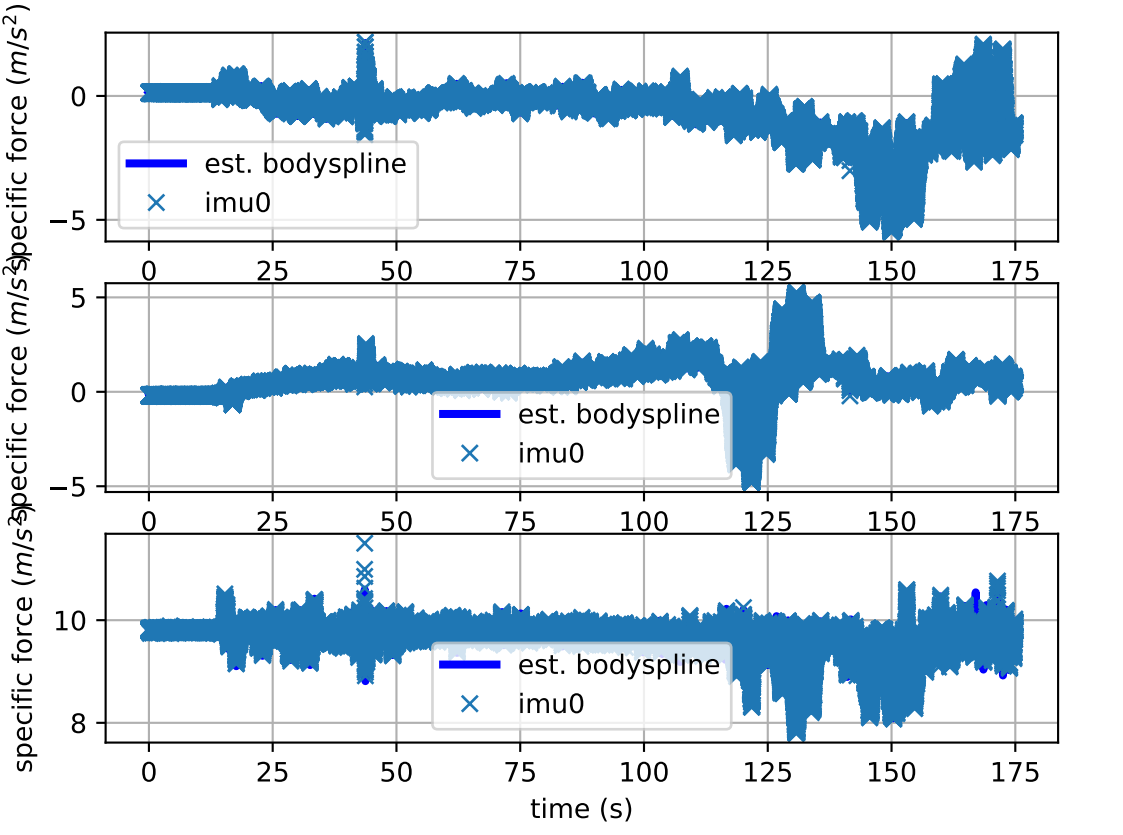
IMU0:

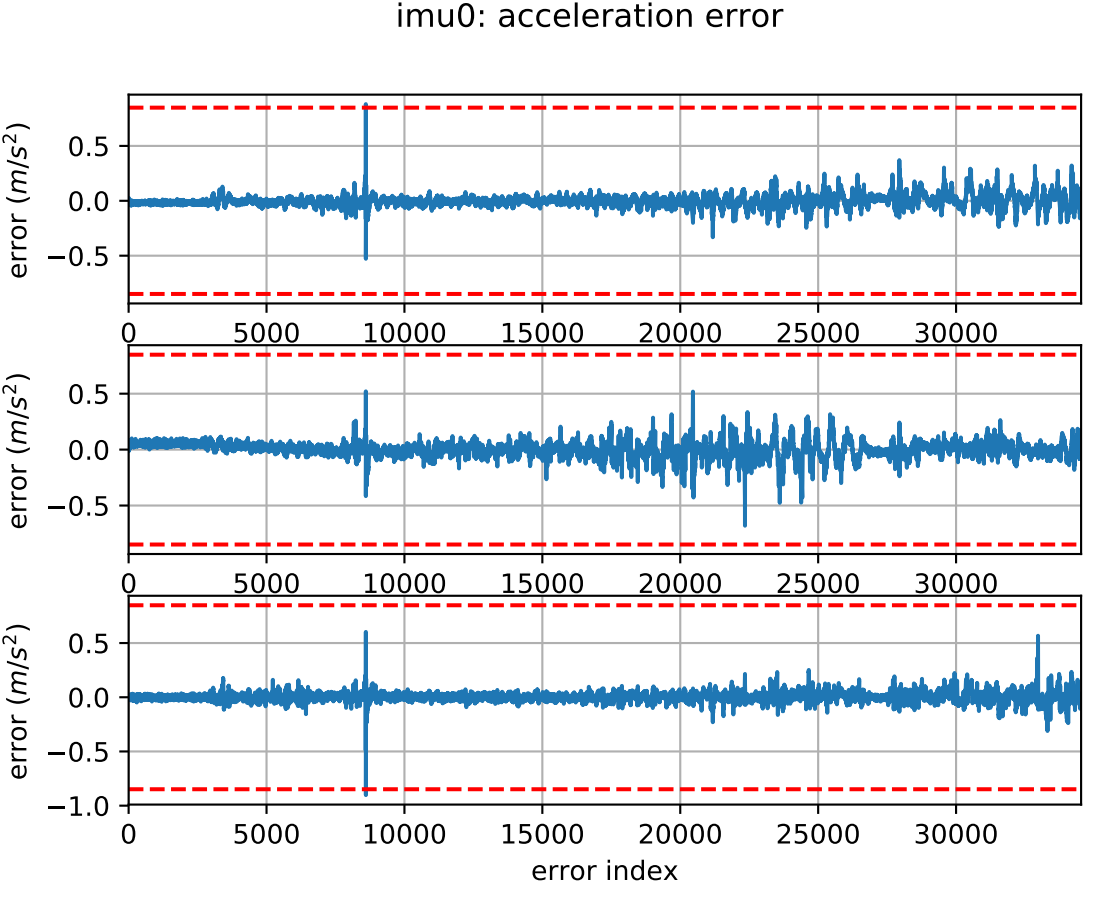
Model: calibrated
Update rate: 200.0

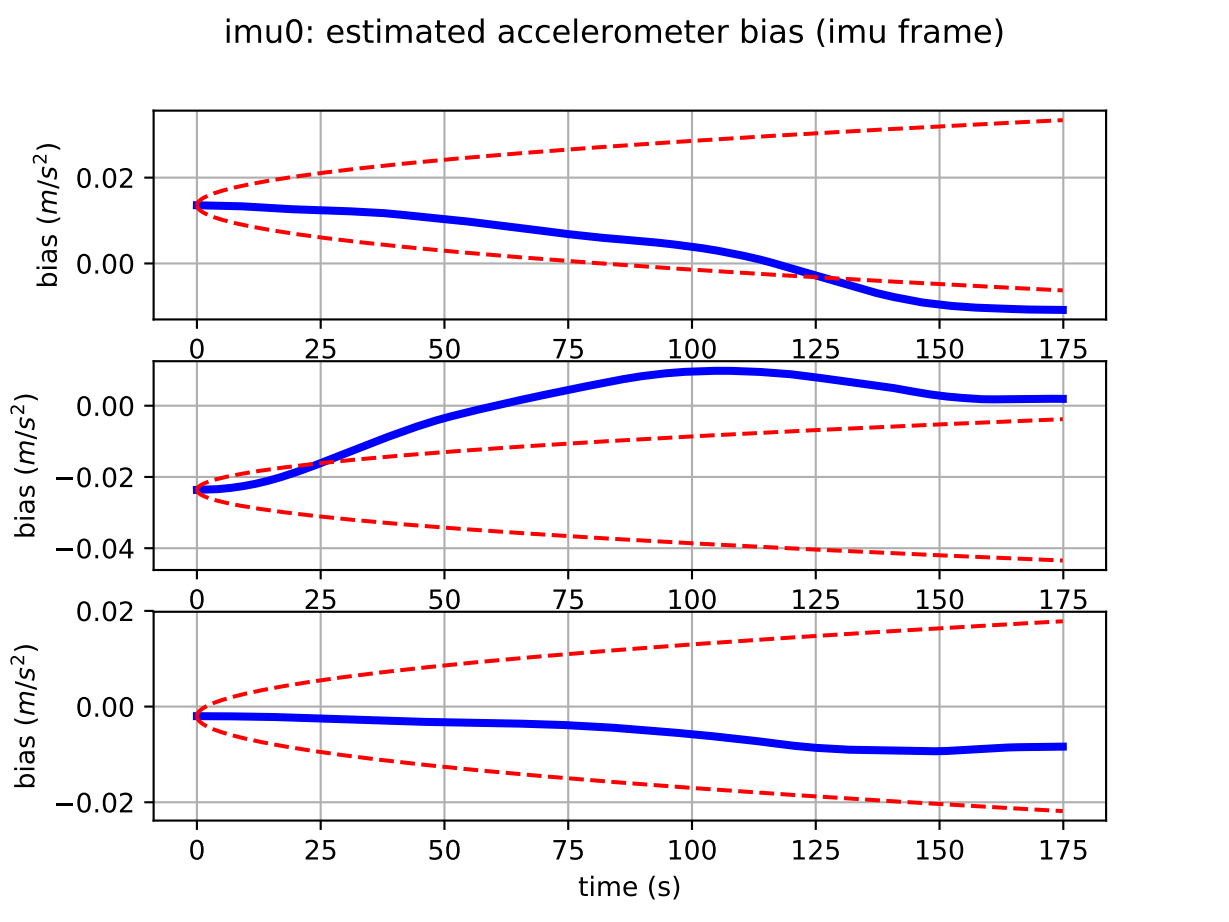
Accelerometer:
Noise density: 0.02
Noise density (discrete): 0.282842712475
Random walk: 0.0005
Gyroscope:
Noise density: 0.002
Noise density (discrete): 0.0282842712475
Random walk: 3e-05
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

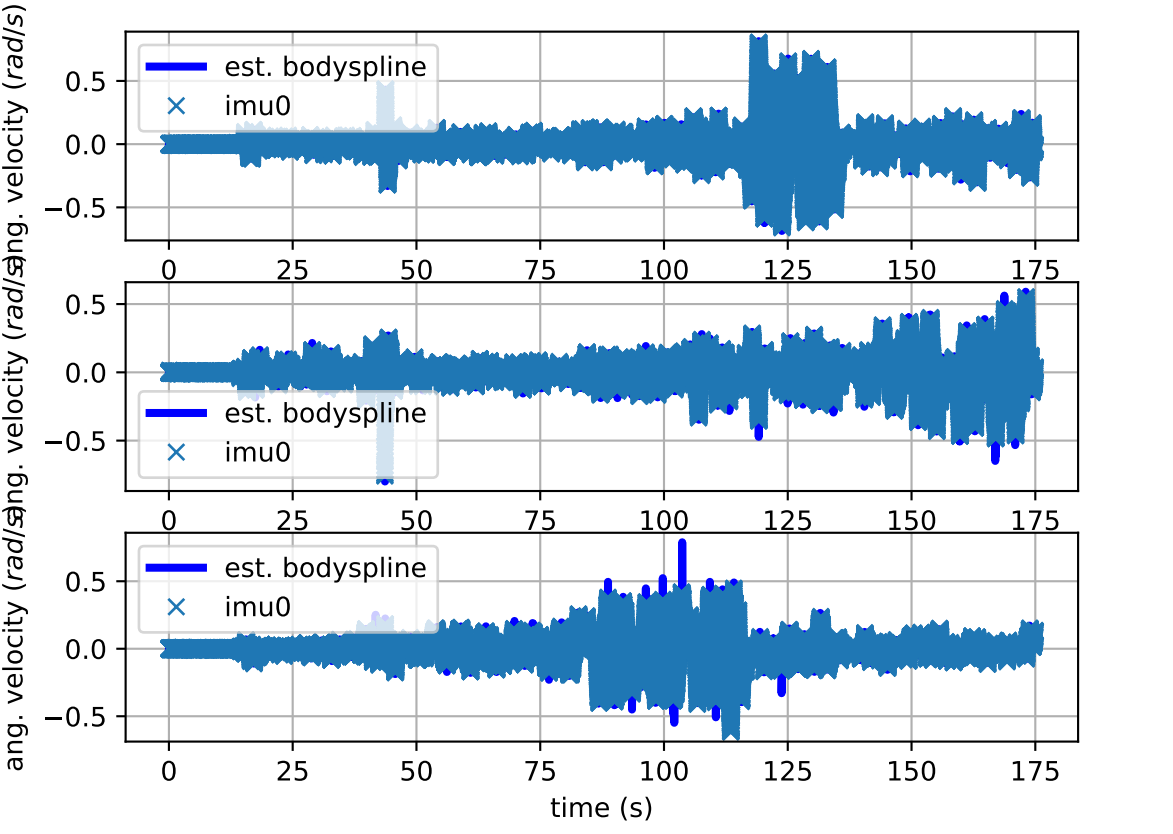

imu0: estimated poses
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
0.1
0.2
0.3
0.4
0.5

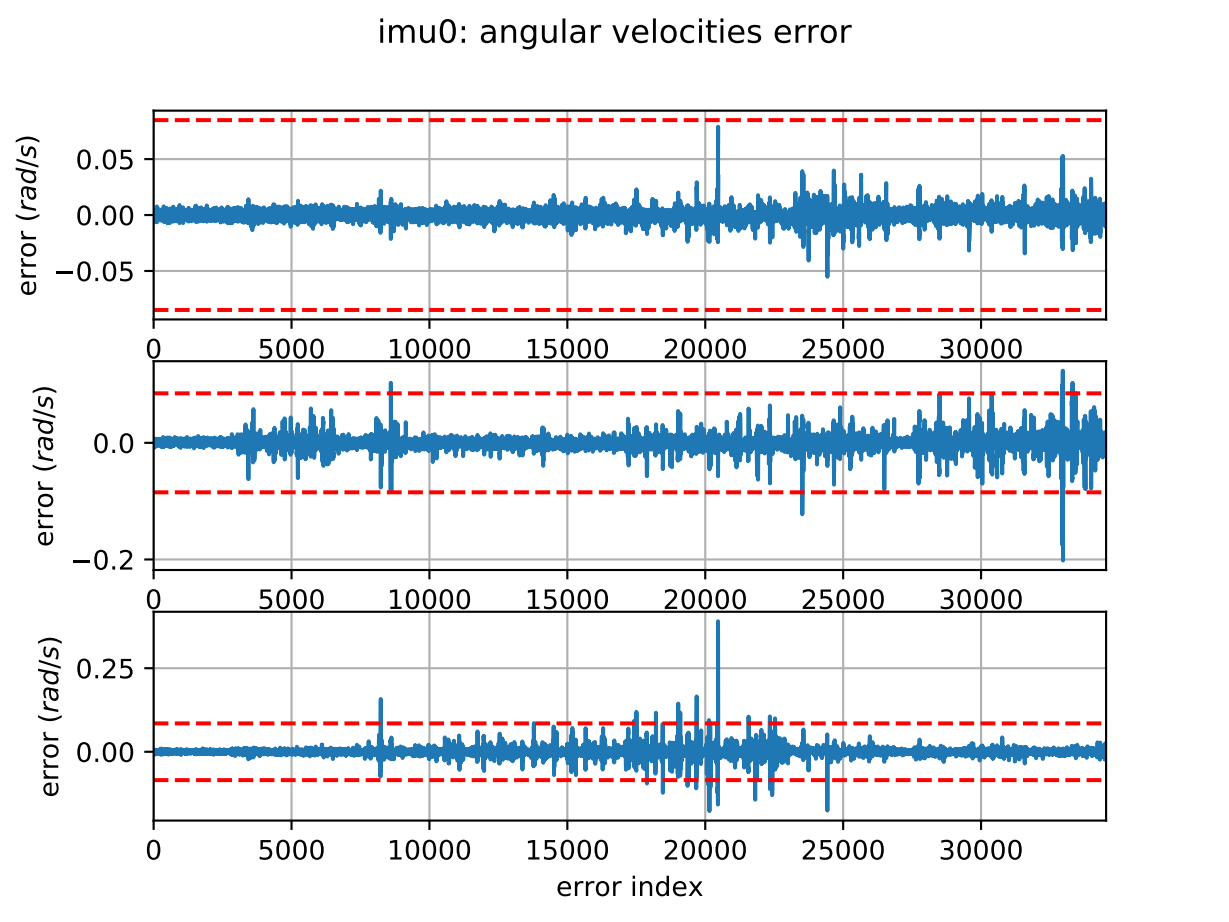

imu0: sample inertial rate
avg dt (ms) = 5.06 +- 0.6049
sample rate (ms)
time (s)
0
2
4
6
8
10
0
20
40
60
80
100
120
140
160

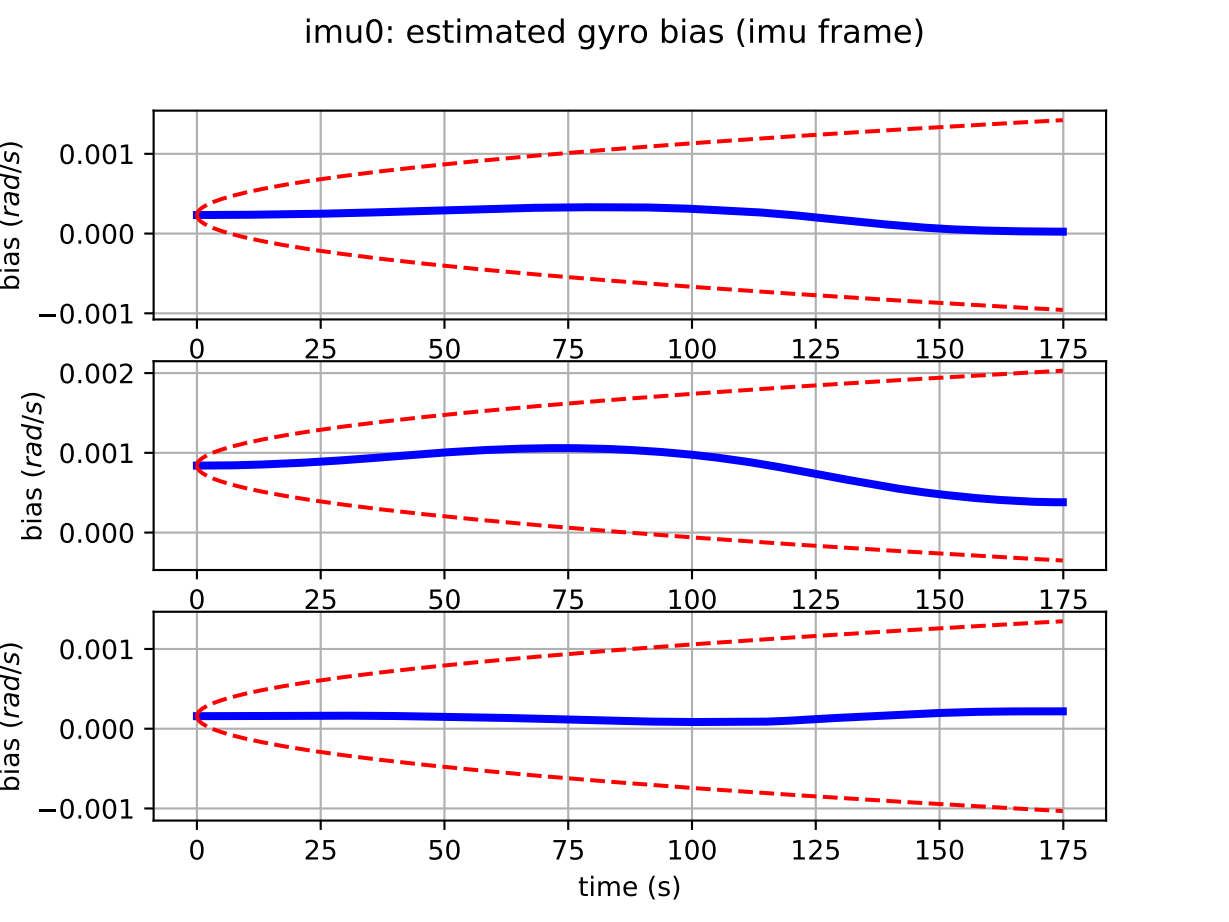

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
0
−5
5
0
−5
10
8
0
25
50
75
100
125
150
175
time (s)

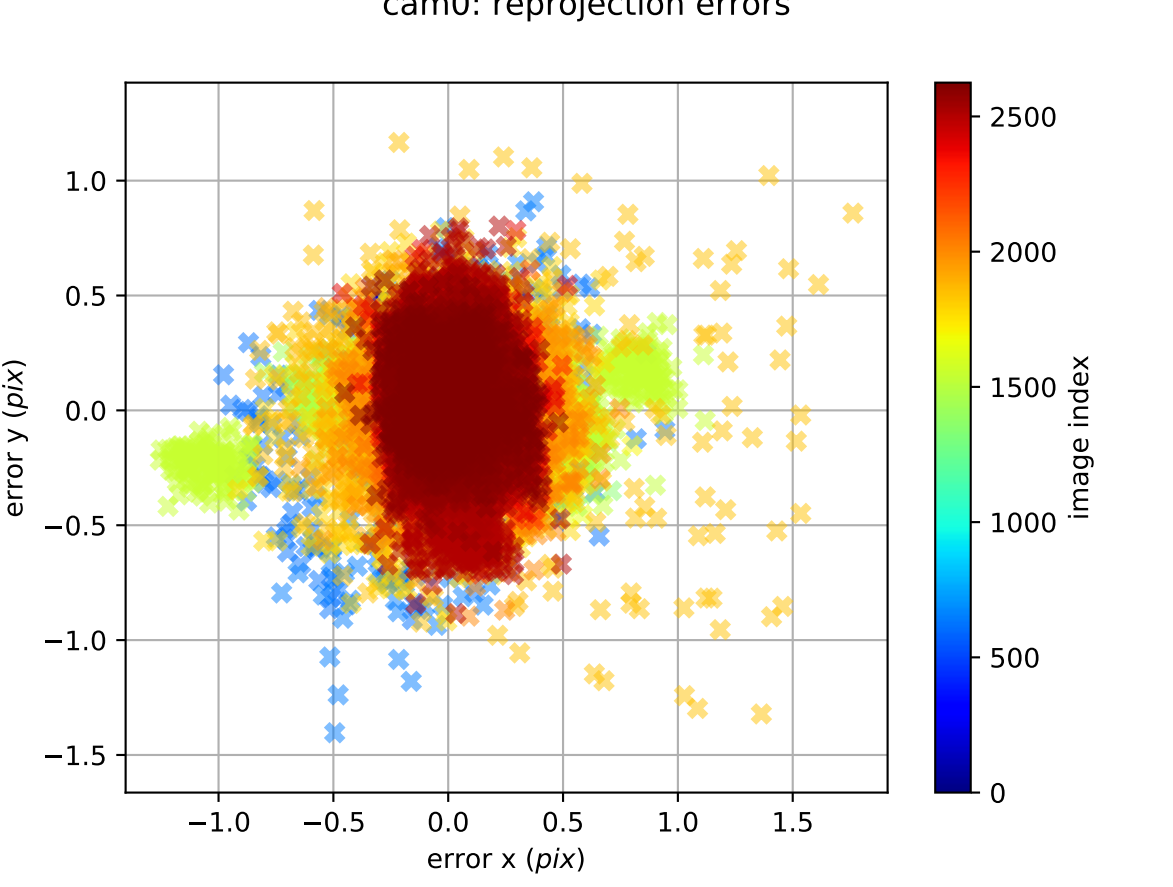

imu0: acceleration error
error (m/s^2)
0.5
0.0
−0.5
0
5000
10000
15000
20000
25000
30000
error (m/s^2)
0.5
0.0
−0.5
0
5000
10000
15000
20000
25000
30000
error (m/s^2)
0.5
0.0
−0.5
−1.0
0
5000
10000
15000
20000
25000
30000
error index

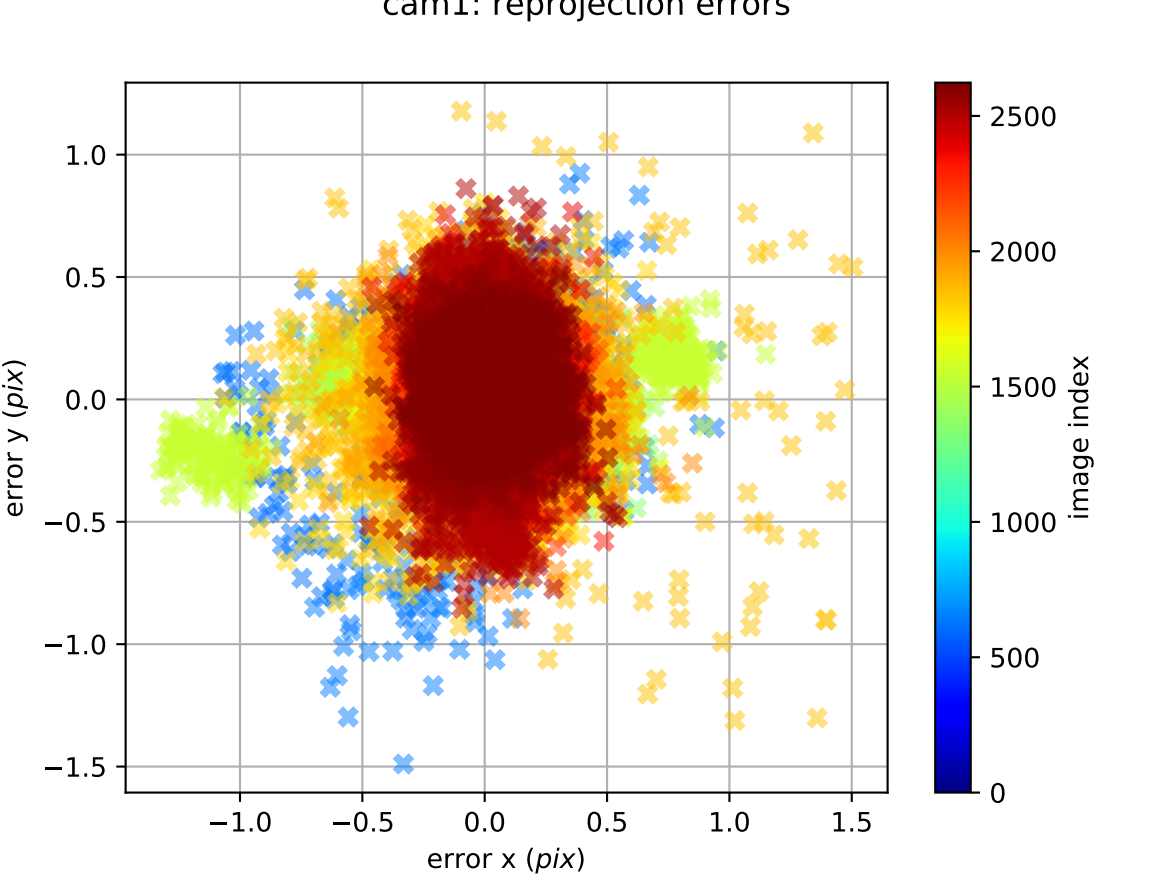

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.02
0.00
0
25
50
75
100
125
150
175
bias (m/s^2)
0.00
−0.02
−0.04
0
25
50
75
100
125
150
175
bias (m/s^2)
0.02
0.00
−0.02
0
25
50
75
100
125
150
175
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
ang. velocity (rad/s)
ang. velocity (rad/s)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
0.5
0.0
−0.5
0
25
50
75
100
125
150
175
time (s)


imu0: angular velocities error
error (rad/s)
0.05
0.00
−0.05
0
5000
10000
15000
20000
25000
30000
error (rad/s)
0.0
−0.2
0
5000
10000
15000
20000
25000
30000
error (rad/s)
0.25
0.00
0
5000
10000
15000
20000
25000
30000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.001
0.000
−0.001
0
25
50
75
100
125
150
175
bias (rad/s)
0.002
0.001
0.000
bias (rad/s)
0.001
0.000
−0.001
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2500
2000
1500
1000
500
0


cam1: reprojection errors
error x (pix)
error y (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.5
1.5
2500
2000
1500
1000
500
0